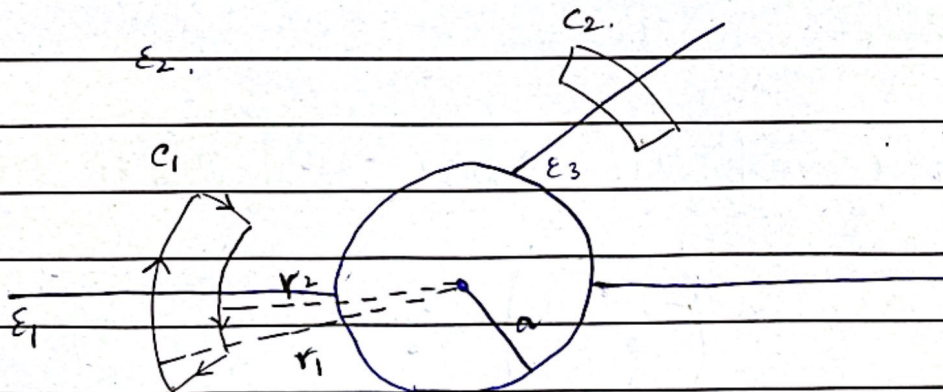
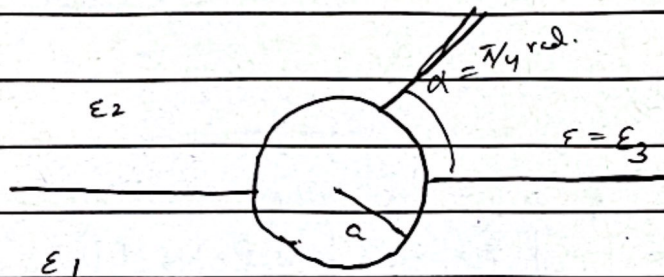


Activity - 4

A conducting sphere of radius a is partially immersed in a liquid dielectric with permittivity ϵ_1 , as shown in the figure. The region above the liquid consists of two gaseous media with permittivities ϵ_2 and ϵ_3 , where medium 3 occupies the angular region bounded by $\alpha = \pi/4$ radians. Given that the total ^{free} charge on the sphere is Q , determine the electric field intensity in all regions.



For C_1 :

$$\int_{r_1}^{r_2} E_{1r} \cdot dr + \int_{r_2}^{r_1} E_{2r} \cdot dr = 0$$

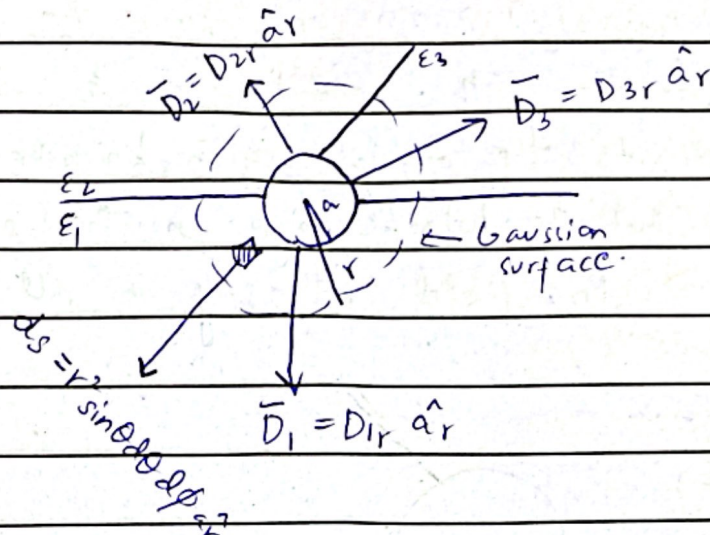
$$\int_{r_1}^{r_2} (E_{1r} - E_{2r}) \cdot dr = 0 \rightarrow \text{Indicates that}$$

$$\bar{E}_{1r} = \bar{E}_{2r}$$

Similarly, for C_2 : $\bar{E}_{2r} = \bar{E}_{3r}$

Hence, $\vec{E}_1 = \vec{E}_2 = \vec{E}_3 = \vec{E}_r$

$\Rightarrow$ Using Gauss's Law ($\oint \vec{D} \cdot d\vec{s} = q_{enc}$) to find what $\vec{E}_r$ is :-



$$\bullet D_{1r} = \epsilon_0 \epsilon_{1r} E_{1r}$$

$$\bullet D_{2r} = \epsilon_0 \epsilon_{2r} E_{2r}$$

$$\bullet D_{3r} = \epsilon_0 \epsilon_{3r} E_{3r}$$

LHS:

$$\int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} \vec{D} \cdot d\vec{s} = \int_{\theta=\pi/2}^{\pi} \int_{\phi=0}^{2\pi} \vec{D}_1 \cdot d\vec{s} + \int_{\theta=0}^{\pi/4} \int_{\phi=0}^{2\pi} \vec{D}_2 \cdot d\vec{s} + \int_{\theta=\pi/4}^{\pi/2} \int_{\phi=0}^{2\pi} \vec{D}_3 \cdot d\vec{s}$$

since, $d\vec{s} = r^2 \sin \theta d\theta d\phi \hat{a}_r$

$$\Rightarrow D_{1r} r^2 (-\cos \theta) \Big|_{\pi/2}^{\pi} \phi \Big|_0^{2\pi} + D_{2r} r^2 (-\cos \theta) \Big|_0^{\pi/4} \phi \Big|_0^{2\pi} + D_{3r} r^2 (-\cos \theta) \Big|_{\pi/4}^{\pi/2} \phi \Big|_0^{2\pi}$$

$$\Rightarrow D_{1r} r^2 2\pi + D_{2r} r^2 2\pi \left(\frac{-\sqrt{2}}{2} + 1 \right) + D_{3r} r^2 2\pi \left(\frac{\sqrt{2}}{2} \right)$$

$$= \left[D_{1r} + D_{2r} \left(1 - \frac{\sqrt{2}}{2} \right) + \frac{\sqrt{2}}{2} D_{3r} \right] r^2 \cdot 2\pi$$

$\rightarrow$ RHS is Q as given.

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$$\left[D_{1r} + D_{2r} \left(1 - \frac{\sqrt{2}}{2} \right) + \frac{\sqrt{2}}{2} D_3 \right] r^2 \pi = Q$$

$$\left[\epsilon_0 \epsilon_{1r} E_{1r} + \epsilon_0 \epsilon_{2r} E_{2r} \left(1 - \frac{\sqrt{2}}{2} \right) + \frac{\sqrt{2}}{2} \epsilon_0 \epsilon_{3r} E_{3r} \right] 2\pi r^2 = Q$$

since $E_{1r} = E_{2r} = E_{3r} = E_r$

$$2\pi r^2 \epsilon_0 E_r \left[\epsilon_{1r} + \epsilon_{2r} \left(1 - \frac{\sqrt{2}}{2} \right) + \frac{\sqrt{2}}{2} \epsilon_3 \right] = Q$$

$$E_r = \frac{Q}{2\pi r^2 \epsilon_0 \left[\epsilon_{1r} + \epsilon_{2r} \left(1 - \frac{\sqrt{2}}{2} \right) + \frac{\sqrt{2}}{2} \epsilon_3 \right]}$$

$$\vec{E}_r = \frac{Q}{2\pi r^2 \epsilon_0 \left[\epsilon_{1r} + \epsilon_{2r} \left(1 - \frac{\sqrt{2}}{2} \right) + \frac{\sqrt{2}}{2} \epsilon_3 \right]} \cdot \hat{a}_r$$